
MA 120: Introduction to Computing System and Databases

Course Code	MA 120
Course Name	Introduction to Computing System and Databases
L-T-P-C	3-0-2-4
Prerequisites	Basic Knowledge of Programming and Data Structures
Intended for	B.Tech Maths and Computing
Distribution	Core
Mutual Exclusion	-
Approval	59th BoA (Apr 2025)

Course Contents

- **Introduction to Computer Organization and Data Representation** [12 Lectures]
 - Data Representation: Number systems (binary, octal, hexadecimal), signed numbers, floating-point representation.
 - Machine Code: Introduction to machine language and assembly language, instruction formats, opcode and operands, addressing modes.
 - Computer Arithmetic: Addition, subtraction, multiplication, and division, fixed-point and floating-point arithmetic, overflow and underflow.
 - Memory Organization and Management: Memory hierarchy, cache design, RAM, ROM, memory addressing, memory paging and segmentation.
- **Introduction to Operating Systems:** [15 Lectures]
 - Responsibilities of Operating Systems: Basic functions of an operating system, user-interface, and system-call interface.
 - Process & Thread Management:
 - * Process model and states: Process life cycle, process control block, context switching.
 - * Thread models: Difference between processes and threads, multi-threading.
 - * User/Kernel Level Threads: Thread management and scheduling at user and kernel levels.
 - Process Creation and Termination: Forking and exec operations, process termination, exit statuses.
 - Process Synchronization: Critical section problem, semaphores, mutex, condition variables.
 - Inter-Process Communication (IPC): Message passing, shared memory, pipes, and sockets.
 - Process Scheduling: Scheduling algorithms such as FCFS, SJF, Round Robin, Priority Scheduling.
 - File Systems: File structure, file access methods, file system implementation, file allocation strategies.
 - • Memory Organization: Primary, secondary memory, memory blocks, paging, segmentation.
 - • Virtual Memory: Address translation, page tables, page faults, thrashing.
- **Introduction to Databases:** [15 Lectures]

- Information Modeling: Conceptual data models, Entity-relationship diagrams, Normal Forms .
- ACID Properties and Transactions: Understanding Atomicity, Consistency, Isolation, Durability, and the role of transactions in databases. Serializability, concurrency control, and lock-based mechanisms like Two-Phase Locking (2PL) and deadlock detection.
- Data Manipulation Language (SQL)
 - * SQL Basics: The basic SQL commands include SELECT, INSERT, UPDATE, and DELETE for query and modifying data.
 - * Advanced SQL: This covers more complex SQL operations such as joins, subqueries, aggregate functions, indexing, and views for efficient data manipulation.

Laboratory Modules

- Based on the contents discussed in class lab will consist of the following:
 - MIPS Programming (QtSPIM)
 - * Writing, running, and debugging MIPS assembly programs.
 - Shell Scripting
 - * Writing and executing bash scripts for task automation.
 - Multiprocessing
 - * Implementing threads, synchronization, and handling race conditions.
 - Multithreading
 - * Implementing threads, synchronization, and handling race conditions.
 - SQL

Text Books

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, **Operating System Concepts**, Wiley.
2. David A. Patterson and John L. Hennessy, **Computer Organization and Design**, Morgan Kaufmann.
3. Raghu Ramakrishnan and Johannes Gehrke, **Database Management Systems**, McGraw-Hill.

References

1. Andrew S. Tanenbaum, **Modern Operating Systems**, Pearson.
2. Hector Garcia-Molina, Jeff Ullman, Jennifer Widom, **Database Systems: The Complete Book**, Pearson.
3. Andrew S. Tanenbaum, **Structured Computer Organization**, Pearson.